



RESEARCH ARTICLE

Plant–soil feedback persists beyond host death to shape density-dependent plant competition

Ching-Lin Huang · Joe Wan · Shuo Wei ·
Chia-Hao Chang-Yang · Po-Ju Ke

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Abstract

Background and Aims Plant–soil feedback (PSF) shapes plant competition, yet classic PSF experiments often overlook the density dependence and temporal complexities of PSFs, especially after host death. We ask whether microbial effects vary after host death and mediate density dependence in seedling competition. **Methods** We combined forest census data from a subtropical forest with a density-gradient greenhouse experiment to evaluate how microbial legacies influence seedling competitive outcome. We used two tree

species (*Engelhardia roxburghiana* and *Machilus zuihoensis*) as model species. In 2023, we collected soil inocula from living and dead trees (died between 2014 and 2019) with unsterilized and sterilized treatments. We applied invasion analysis to infer seedling competitive outcomes and used bootstrapping to evaluate uncertainty.

Results Soil microbial communities shaped seedling competitive outcome, favoring *E. roxburghiana*. We found a consistent exclusion of *M. zuihoensis* by *E. roxburghiana* when using unsterilized soils collected from living and dead hosts, indicating that PSF persists for over 4–9 years after host death. In contrast, we found *E. roxburghiana* persisted in sterilized inoculum from living but not from dead hosts, suggesting underlying abiotic changes which might be masked by microbial effects. Moreover, we found significant evidence that soil microbes can mediate negative, nonlinear density dependence in seedling competition.

Conclusion We provide experimental evidence of persistent microbial legacies that shape plant competition in a subtropical forest. Our study demonstrates how integrating field-based census data with density-gradient experiments and explicit uncertainty estimation can better capture the temporal dimensions, complexities of density dependence, and uncertainties of PSF.

Keywords Coexistence · Forest dynamic plot · Invasion growth rate · Legacy effect · Nonlinear density dependence · Subtropical forest

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Ching-Lin Huang and Joe Wan contributed equally to this work.

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C.-L. Huang · J. Wan · S. Wei · P.-J. Ke (✉)
Institute of Ecology and Evolutionary Biology, National Taiwan University, Taipei, Taiwan
e-mail: pojuke@ntu.edu.tw

C.-L. Huang
Department of Ecology, Evolution and Behavior, University of Minnesota-Twin Cities, St. Paul, MN, USA

S. Wei
Yale School of the Environment, Yale University, New Haven, CT, USA

C.-H. Chang-Yang
Department of Biological Sciences, National Sun Yat-sen University, Kaohsiung, Taiwan

Introduction

Reciprocal interactions between plants and soil microbes, known as plant–soil feedback (PSF), shape the structure and dynamics of plant communities (Bever et al. 1997, 2010, van der Putten et al. 2013). Plants, as hosts, condition the soil microbial community differently, altering the relative abundance of beneficial and detrimental soil microbes among host plants (Bever et al. 2012, Hu et al. 2018). In turn, these shifts in plant species-specific soil microbial composition influence the growth and survival of nearby plant individuals and the establishment of new recruits (Bever 2003, Bever et al. 1997). As plants vary in their responses to soil microbial communities, PSF can alter competitive dynamics by modifying the relative competitive abilities among plants (Kandlikar et al. 2019, Ke and Wan 2020), thereby shaping their relative abundance patterns (Klironomos 2002, Mangan et al. 2010) and assembly dynamics (Fukami and Nakajima 2013). Despite the widely recognized importance of PSF for plant community dynamics, most empirical studies quantify PSF by growing a single plant individual in soils collected at a single time point, leaving the density dependence and temporal complexity of PSF underexplored (Chung 2023, Gundale and Kardol 2021, Ke et al. 2025). These complexities are expected since soil microbes respond to host density and turn over through time (Steinauer et al. 2023, Willing et al. 2024). Resolving the density- and time-dependencies of PSF is essential for explaining and predicting plant coexistence and community dynamics.

To understand how soil microbes influence plant competitive outcome, studies estimate indices derived from plant–soil feedback theory by growing plants in soils conditioned by different plant species (Bever et al. 1997, Crawford et al. 2019, Yan et al. 2022). However, these experiments typically measure the responses of individual plants in isolation, implicitly assuming that soil microbes only modify the intrinsic growth rate of plant populations (Bever et al. 1997, Kandlikar 2024). This classical assumption has been challenged by experiments showing that microbial effects can vary with plant density, indicating density dependence in plant–soil microbe interactions (West 1996, Willing et al. 2024). Moreover, this approach overlooks the long-standing recognition that plant–plant interactions are inherently density-dependent; that is, plant performance is influenced by the abundance of neighboring

plants (Watkinson and Freckleton 1996). Indeed, growing evidence indicates that density-dependent plant competition and soil microbial effects combine to determine overall competitive dynamics in plant communities (Tao et al. 2026, Wasan and Bennett 2026). While some recent studies have employed various experimental designs to quantify how soil microbes mediate the density dependence of plant–plant interaction (e.g., Cardinaux et al. 2018, Chung and Rudgers 2016, Huangfu et al. 2022), we lack practical guidelines to address this aspect of PSF. To this end, Ke and Wan (2023) proposed a density-gradient experimental design to measure density-dependent microbial effects on plant performance, which can also detect nonlinearity in density dependence. Building upon previous work linking plant–soil feedback to coexistence (Ke and Wan 2020), the method quantifies the invasion growth rate (IGR) as a theoretically rigorous metric for a species' ability to persist under competition (Grainger et al. 2019, Turelli 1978). This framework offers a tractable approach for assessing how soil microbes influence plant competitive outcomes by altering the form of density-dependent plant–plant interactions.

Recent studies have also underscored the importance of the temporal complexity of plant–soil microbe interactions (Chung 2023, Gundale and Kardol 2021, Kardol et al. 2013, Ke et al. 2025). Specifically, to better predict the impact of conditioned microbial communities on newly arriving seedlings, it is essential to examine how microbial effects develop over a conditioning plant's lifespan and how they vary after host death. Theoretical studies have shown that persistent microbial legacies are crucial for PSF to maintain plant diversity (Ke et al. 2021, Miller and Allesina 2021); however, empirical studies have yielded contrasting results regarding the persistence of these microbial legacies. For instance, Bennett et al. (2022) used field soils from living or dead tree hosts to show strong and consistent negative legacy effects on seedling growth and survival. However, Ou et al. (2024) demonstrated that a six-month temporal lag between conditioning and planting can reverse the competitive hierarchy among species, suggesting that microbial effects are not constant after host mortality. Moreover, while a recent study using forest census data suggests that density-dependent plant–plant interactions are persistent after tree death and implicates a potential role for soil microbes (Magee et al. 2024), experimental studies that directly test the microbial mechanisms underlying such patterns are still lacking.

Here, we explore how density-dependent plant–plant interactions are influenced by soil microbial legacies left by dead host plants, testing the hypothesis that the effects of soil microbes on plant competitive outcome vary through time. To address this, we combine forest census data with a density-gradient greenhouse experiment (modified from Ke and Wan 2023), using soils collected beneath living and dead trees to inoculate a seedling competition experiment. We leverage a well-studied field system: Fushan Forest Dynamic Plot (Fushan FDP), a subtropical forest in Taiwan with four complete forest censuses. Given that preparing soils representing multiple years since host death is often infeasible in greenhouse settings, we use field soil beneath living and dead trees to study the temporal changes of PSF by leveraging different host tree statuses in nature. Our experiment monitors seedling competition under a density-gradient experimental design to estimate population-level intrinsic growth rates and density-dependent plant–plant interactions, either with or without microbial mediation. Finally, we apply invasion analysis to evaluate species persistence and competitive outcomes, and compare how these outcomes shift across microbial treatments and over time.

Materials and methods

Study species and field soil preparation

We studied a pair of common subtropical low-montane forest canopy trees, *Engelhardia roxburghiana* (Juglandaceae) and *Machilus zuihoensis* (Lauraceae), where canopy tree species were defined as those with individuals reaching more than 50 cm diameter at breast height (Su et al. 2007). Both are among the most abundant species at Fushan Forest Dynamics Plot (Fushan FDP; 24°45′40″ N, 121°33′28″ E; elevation 600–733 m a.s.l.). Namely, *E. roxburghiana* is the most frequent canopy tree species by number of individuals (mean $\pm$ 2 SE = 270.6 ± 36.6 individuals/ha; 17.0% of all canopy-tree individuals) and *M. zuihoensis* is the 7th most frequent (100.2 ± 11.3 individuals/ha; 6.3% of all canopy-tree individuals). In terms of above-ground biomass, *M. zuihoensis* ranks 4th (15.0 ± 1.2 Mg/ha), while *E. roxburghiana* ranks 8th (6.2 ± 0.3 Mg/ha) (Su et al. 2007). From 2004 to 2024, the annual mortality rate was approximately $3.6\% \pm 0.11\%$ for *E. roxburghiana* and $2.1\% \pm 0.14\%$ for *M. zuihoensis*, mainly

due to crown damage (Zuleta et al. 2022). While both study species are large canopy trees, they are likely to interact differently with soil microbes. For instance, *E. roxburghiana* is capable of forming ectomycorrhizae (Haug et al. 1994), while *M. zuihoensis* associates only with arbuscular mycorrhizal fungi. Moreover, they differ in their phylogenetic relatedness to surrounding vegetation: *E. roxburghiana* is the only member of the Juglandaceae at the site, whereas *M. zuihoensis* is one of nearly a dozen species in the Lauraceae at Fushan FDP (including 4 out of the 10 most common canopy trees).

Field soils were collected from the Fushan FDP in late November 2023. For both species, we randomly selected 10 individuals in each of two statuses: living (individuals that were alive at the time of sampling) and dead (individuals that had died between 2014 and 2019, i.e., had been dead for 4 to 9 years, according to forest census data; Fig. S1). To ensure that dead individuals could be accurately located in the field and control for tree age, we restricted our sampling to relatively mature individuals in the 2019 tree census (diameter at breast height 14.5–38 cm; average: 24.5 cm). Note that only nine living *M. zuihoensis* individuals were sampled, as one individual was found dead during sampling. Using a surface-sterilized soil core sampler (0.5% bleach followed by 75% ethanol), we collected 1.0 L of top-soil (~10 cm depth, excluding litter), pooling several cores per selected individual. Given the high density of sapling and adult tree individuals at Fushan FDP (Su et al. 2007), we restricted our soil sampling to a 1 m radius around each individual to limit the influence of neighboring vegetation. Soils from different individuals were kept in separate sealed sterile Ziploc bags, transported to the lab in a cooler, sieved (2 mm mesh) to remove stones and root fragments, and stored at 4 °C before being used in the greenhouse experiment.

A mixed soil sampling approach (Gundale et al. 2017) was used to prepare the soil inoculum for our greenhouse experiment. Specifically, we mixed and homogenized equal volumes of soil collected from all individuals of the same species and status (living or dead). This soil mixture was then mixed with autoclaved perlite following a 6:1 soil:perlite volume ratio to improve drainage, forming the inocula used in our greenhouse experiment. As a baseline for assessing the potential effects of soil microbes, we sterilized half of the mixed soil inocula by autoclaving (121 °C and 1.2 kg/cm² for 30 min, incubation at room temperature for

24 h, and another 121°C and 1.2 kg/cm² for 30 min). Overall, for each species, our experiment included four different inoculum types, defined by the two conditioning host plant statuses (living or dead based on forest census) and the two sterilization treatments (unsterilized or sterilized).

Density-gradient greenhouse experiment

Our greenhouse experiment was conducted according to the following procedure. We germinated seedlings of *M. zuihoensis* and *E. roxburghiana* in growth chambers. Fresh seeds, sourced from central Taiwanese forests via a commercial seed supplier, were surface-sterilized with 0.5% bleach solution and rinsed with reverse osmosis water before germination. Since the seeds of *M. zuihoensis* experience physiological dormancy (Chien et al. 1994), the seeds were cold stratified at 4°C for two months in sealed sterile Ziploc bags filled with sterilized sphagnum moss. In October 2023, once the *M. zuihoensis* seeds started to germinate, seedlings with extended primary leaves were transferred to trays (8 cm in depth) filled with sterilized perlite to avoid root girdling. Meanwhile, we directly sowed *E. roxburghiana* seeds on identical trays as *M. zuihoensis*. Both species were placed in growth chambers maintained at 25/20°C day/night temperature, 12/12 h day/night photoperiod (light intensity $\sim 150 \mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$), and a constant 90% relative humidity. Seeds and seedlings were misted daily with reverse osmosis water to maintain substrate moisture.

Following the experimental setup proposed in Ke and Wan (2023), we implemented a full factorial design consisting of the two plant focal species $\times$ two plant competitors $\times$ four competitor density scenarios $\times$ two field host plant statuses (living or dead) $\times$ two soil sterilization treatments (unsterilized or sterilized), with four replicate pots for each combination (a total of 256 pots, Fig. 1a). In December 2023, similarly sized seedlings of *M. zuihoensis* and *E. roxburghiana* were transplanted into pots (10.6 cm in top diameter, 16.0 cm in height, ~ 1 L in volume). Each pot contained one focal individual along with a designated number (0, 1, 2, or 4) of conspecific or heterospecific competitors. The pots were filled with 0.8 L of sterilized planting substrate (peat:perlite = 7:1, by volume), then with 0.1 L of the designated soil inoculum, and finally topped with another 0.1 L of sterilized planting sub-

strate. As this experiment is designed to quantify the growth of each species when it is rare (i.e., invasion growth), the inoculum always matched the competitor species. For example, if the competitor was *M. zuihoensis*, we used the inoculum collected from *M. zuihoensis* in the field. The setup thereby simulates the microbial effects conditioned by a common competitor species (i.e., the resident) as part of its impact on the rare focal species (i.e., the invader). To ensure future identification, we marked every focal individual with a 5 cm plastic stirrer. The whole planting process was completed within three days. Immediately after planting, 15 additional seedlings of each species from the germination trays were oven-dried at 70°C for 72 hours to measure their average initial dry biomass. To minimize transplant shock, we began the experiment in growth chambers under identical settings to those used for seed germination. After 80 days, as plants began to fill the growth chamber space, we transferred all pots to the phytotron at National Taiwan University under 25/20°C day/night temperature and a natural photoperiod. Throughout the experiment, plants were watered with 30–50 mL of reverse osmosis water two to three times per week, and were regularly weeded to minimize the effects of non-target species on seedling performance.

After six months, we harvested the above- and below-ground tissues of each individual plant between late June and early July 2024. We gently washed all soil from root systems and separated the above- and below-ground tissues based on the position of the highest secondary root. Entangled fine roots from different individuals were carefully separated. We retained as many root fragments as possible and weighed them separately (see “Data analysis” section for how root fragments contribute to calculating individual biomass). Finally, we oven-dried above- and below-ground tissues at 70°C for at least 72 hours and summed their weights to determine total dry biomass per focal individual. We observed 21 dead individuals, including twelve competitors and nine focal individuals, out of 704 individuals (overall mortality rate: 3%). The above- and below-ground tissues of these dead seedlings were harvested concurrently at the end of the experiment using the same method as for the surviving ones. We used these dead seedling measurements in subsequent analyses, an approach which implicitly incorporates mortality by accounting for dead seedlings’ inability to accumulate biomass.

Data analysis

To examine how plant competitive outcome is affected by soil microbes, we first analyzed how focal species' growth responded to competitor biomass in different inocula using independent second-order polynomial regressions. In these regressions, the intercept term estimates the focal species' growth in the absence of competitors, whereas the linear and quadratic terms estimate how increasing competitor biomass affects focal species performance, that is, the overall density-dependent effect of competitors on focal species' growth. Second, based on these growth regressions, we assessed species persistence and coexistence under different inoculum types using invasion analysis. This assumes that biomass accumulation is a proxy for long-term fitness (Younginger et al. 2017), reflecting findings from Fushan FDP showing that early-stage growth rate is one of the most important bottlenecks in the tree life cycle (Chang-Yang et al. 2021). Third, we evaluated the uncertainty in model parameters and competitive outcomes by bootstrapping. Finally, based on bootstrap distributions, we compared the differences in competitive outcomes between unsterilized versus sterilized inocula and soils from living versus dead hosts to infer how microbial effects on plant competition outcomes change over time. Here, because growth and mortality of competitors varied among pots, we used a dimensionless $\log(1 + x)$ transformation of competitor biomass at harvest as a proxy for competition intensity (Baiocchi 2012, Cardinaux et al. 2018, Reader et al. 1994) and report its effects on focal species' growth as a measure of realized competitive effect. Nevertheless, results were qualitatively similar when we repeated all analyses using competitor density or $\log(1 + x)$ transformation of competitor biomass in grams as the predictor (see Figs. S3 and S4). Detailed methods for each step are as follows.

First, we analyzed how each focal species' growth responded to competitor biomass in each inoculum type. For each combination of competitor species, field host status (living versus dead), and inoculum sterilization (unsterilized versus sterilized), we fit an independent second-order polynomial regression.

We defined the growth of a focal individual as the log-ratio of its harvested dry biomass to the species' average initial dry biomass, serving as a proxy of the species' per-capita population growth rate. This focal individual growth was regressed against competitor

harvested dry biomass independently for each combination of competitor species, field host status, and inoculum sterilization treatment. Specifically, we considered a second-order polynomial regression model $y = r + \alpha x + \beta x^2 + \epsilon$, where y represents the focal individual growth, and x is the log-transformed, dimensionless harvested dry biomass of the competitors, defined as $\log(1 + \text{biomass}_{\text{harvest}}/\text{biomass}_{\text{initial}})$. We used a $\log(1 + x)$ transformation of competitor biomass to improve homogeneity of variance. Competitor biomass across all density treatments was scaled by the species-specific average initial biomass of a single seedling to obtain a dimensionless value and to avoid sensitivity of the transformation to measurement units (Baiocchi 2012). The intercept r represents the growth of the focal species in the absence of competitors. The coefficients α and β represent the linear and quadratic competitor biomass effects on the focal species' growth. These effects can be conspecific or heterospecific depending on whether the focal and competitor species are the same. The error term ϵ was assumed to follow a normal distribution. To select better-fitted models, we conducted F-tests to compare across nested models and retained the quadratic term only when it significantly improved model fit (AIC and BIC values shown in the Supplementary Information). Theoretical studies have highlighted the importance of the ratio between linear conspecific and heterospecific competitive effects for inferring competitive outcomes (Adler et al. 2018, Ke and Wan 2023). As our main goal was to investigate competitive outcomes across different inocula, we kept the linear terms in all models regardless of their significance (Van Dyke et al. 2024) and assessed their uncertainty using bootstrapping, as described below.

Second, we conducted invasion analyses following Ke and Wan (2023) to determine the competitive outcomes between *M. zuihoensis* and *E. roxburghiana* across inocula differing in host statuses and sterilization treatments. This approach estimates the sign and relative magnitude of each species' invasion growth rate (IGR), which is the population growth rate of a focal species when its competitor (the resident) has reached its monoculture equilibrium. To obtain the monoculture equilibrium for each species, we extrapolated the fitted regression model to determine the *conspecific* competitor biomass at which growth of the species equals zero, i.e., the non-negative x-intercept of the fitted growth function (vertical dotted blue line in Fig. 1b; illustrated for the nonlinear case when the quadratic term β is

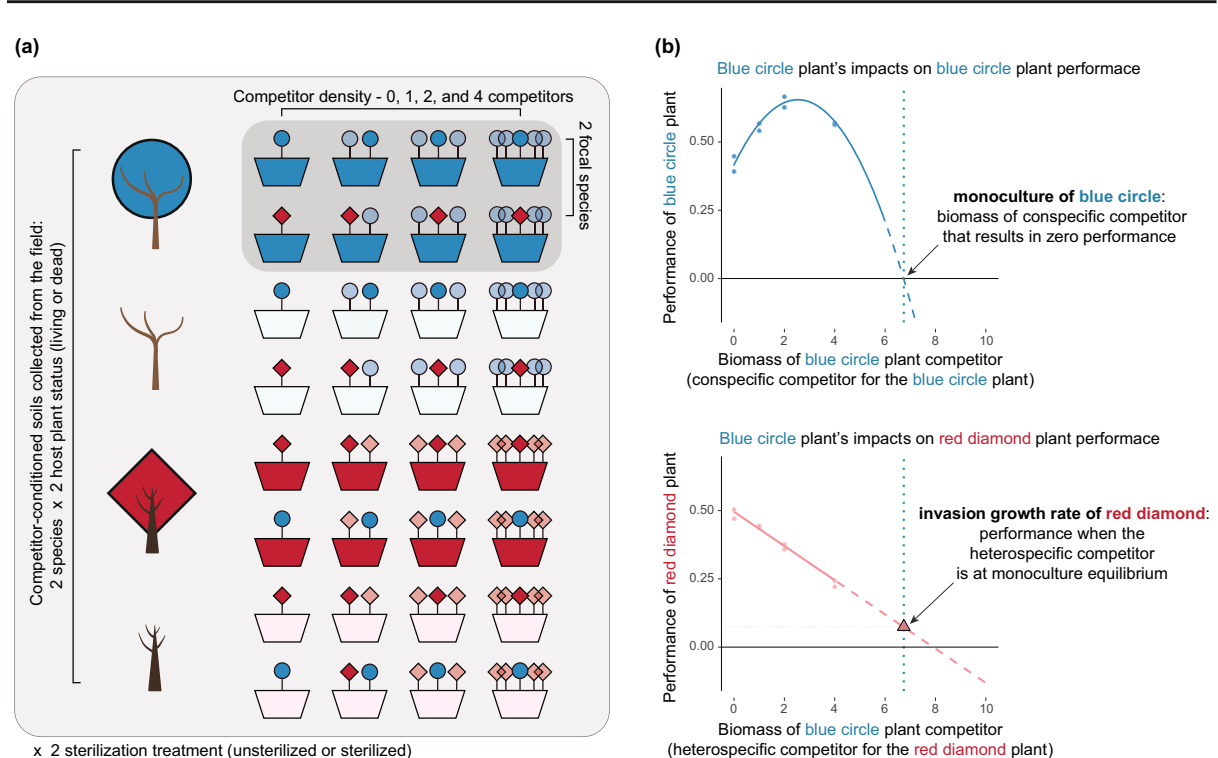


Fig. 1 Experimental design and schematic diagram for the invasion analysis following Ke and Wan (2023). **(a)** The experiment measures the performance (growth) of a focal individual (fully opaque blue circle or red diamond plants, representing the two tree species considered in our study) competing against different densities of competitors (translucent individuals). We added competitor-conditioned soils collected from either living or dead tree individuals in the field (inocula from different tree statuses are represented as pots with opaque and translucent colors, respectively; see also tree icons on the left representing host tree species and status). The depicted setup was repeated for two sterilization treatments (i.e., either unsterilized or sterilized inoculum) and was replicated four times, resulting in a total of 256 pots. **(b)** An example of how to quantify the invasion growth

rate for the red diamond plant, using pots within the highlighted gray box in (a). Both diagrams show a regression model relating focal species growth to competitor biomass: the blue circle plant shows a nonlinear density-dependent relationship, whereas the red diamond plant shows a linear relationship. The dashed portion indicates the extrapolated range. First, the impact of the blue circle plant (the competitor, with biomass drawn on the x-axis) on conspecific performance (upper panel) provides an estimation of its monoculture equilibrium (vertical blue dotted lines). Second, the performance of the red diamond plant (lower panel) when the blue circle plant is at its monoculture equilibrium represents the red diamond plant's invasion growth rate (red triangle). Note that this approach applies regardless of whether the fitted model is linear or includes the nonlinear quadratic term

retained). As a proxy for a focal species' IGR, we used the fitted model to predict the focal species' growth when its *heterospecific* competitor was at monoculture equilibrium (red triangle in the lower panel of Fig. 1b). A special case can arise when the linear conspecific competitor effect is positive and no nonlinear term is retained during model selection. In this case, the species' growth is predicted to increase without bound as conspecific competitor biomass increases, indicating that the competitor biomass required to generate competitive limitation lies beyond our experimental design and the species' monoculture equilibrium can-

not be determined. When the heterospecific competitor's monoculture equilibrium could not be determined, we assigned the sign of the focal species' IGR based on the sign of the highest-order heterospecific effect included in the model. Although the exact magnitude of the IGR cannot be determined in such cases, the sign of the IGR alone is sufficient to infer species persistence and competitive outcomes. Therefore, for each inoculum, we qualitatively inferred species persistence and overall competitive outcome from the pair of IGR signs. A positive IGR indicates that the species can persist under competition, whereas a negative IGR suggests

that the species would be outcompeted. If both species have positive IGRs, they are predicted to coexist; if only one species has positive IGR, that species is predicted to exclude its competitor; if both species have negative IGRs, they are inferred to show priority effects, where the outcome depends on the initial abundances of the two species.

Third, we assessed the robustness of the invasion analysis results by non-parametric bootstrapping. This allowed us to account for uncertainty in the estimated competitive outcomes, as recently suggested by Terry and Armitage (2024). Specifically, within each combination of focal species, competitor species, host status, and sterilization treatment, we resampled the 16 data points with replacement, consisting of four competitor densities with four replicates, while maintaining the original sample size. Using the same model structure identified during model selection, we re-estimated model parameters (i.e., the growth without competitors r , linear competitor effect α , and quadratic competitor effect β). We then repeated the invasion analysis for each bootstrap sample and recorded the resulting competitive outcomes. This procedure was repeated 10,000 times to generate the bootstrap distributions for the model parameters and competitive outcomes. The 90% confidence intervals for all model parameters were calculated using the percentile method (Davison and Hinkley 1997). Significant differences in model parameters across inocula were determined by non-overlapping confidence intervals. Furthermore, we used the bootstrap distributions to evaluate the robustness of the inferred competitive outcomes. A positive IGR was regarded as a robust inference if $> 95\%$ of bootstrap samples yielded a positive value (i.e., a one-sided test at the $\alpha = 0.05$ significance level), suggesting strong evidence for persistence. Similarly, we regarded a competitive outcome as robust if it received over 95% support across the bootstrap replicates. Finally, to infer microbial effects on seedling competition, we compared competitive outcomes between unsterilized versus sterilized inocula and soils collected from living versus dead hosts. Here, given there is no established convention for significance testing of competitive outcomes (Terry and Armitage 2024), we follow recent practice (Van Dyke et al. 2024, Willing et al. 2024) by comparing proportions of bootstrap support for each competitive outcome to convey the robustness of our inferences.

Results

Significance and uncertainties of model parameters

Our model selection process identified a significantly negative quadratic term only in the case when *E. roxburghiana* (focal species) competed with *M. zuihoensis* (competitor) in sterilized inoculum collected under dead hosts ($F = 8.279$, $P = 0.013$; Table S1). Model comparisons using competitor density instead of biomass as the explanatory variable did not show systematic improvements in model fit (Table S2). Hence, we focus our discussion here on the models using competitor biomass (Fig. 2) and present the results based on competitor density in the Supplementary Information (Figs. S2 and S3).

Based on bootstrap samples, we found *E. roxburghiana* generally had greater single individual growth than *M. zuihoensis* across most inocula (comparison between blue and red bars in Fig. 3a). For *M. zuihoensis*, there was no significant difference (i.e., overlapping bootstrap CI) in growth when it grew alone in different inocula. In contrast, *E. roxburghiana* showed a significantly higher single individual growth in unsterilized inocula from living conspecific hosts (dark blue bar; estimate = 3.208, 90% CI: 2.735 – 3.834; Table S3), compared to that in sterilized inocula from living conspecific hosts (estimate = 2.253, 90% CI: 1.196 – 2.461) and in unsterilized inocula from dead conspecific hosts (estimate = 1.826, 90% CI: 1.522 – 2.122).

Most of the competitor effects were not significantly different from zero. Two exceptions stood out: (1) the significantly negative quadratic heterospecific competitor effect on *E. roxburghiana* (light blue bar in Fig. 3b; estimate = -0.736 , CI: -1.166 – -0.299 ; Table S4) in sterilized inocula from dead hosts, and (2) a significantly positive linear conspecific competitor effect among *E. roxburghiana* (dark blue bar; estimate = 0.212, CI: 0.105 – 0.312) in unsterilized inocula from dead hosts, which also differed significantly from the corresponding effect in sterilized inocula (estimate = -0.169 , CI: -0.430 – 0.038).

Robustness of species persistence and competitive outcomes

Invasion analysis based on the selected fitted models revealed the estimated species persistence and compet-

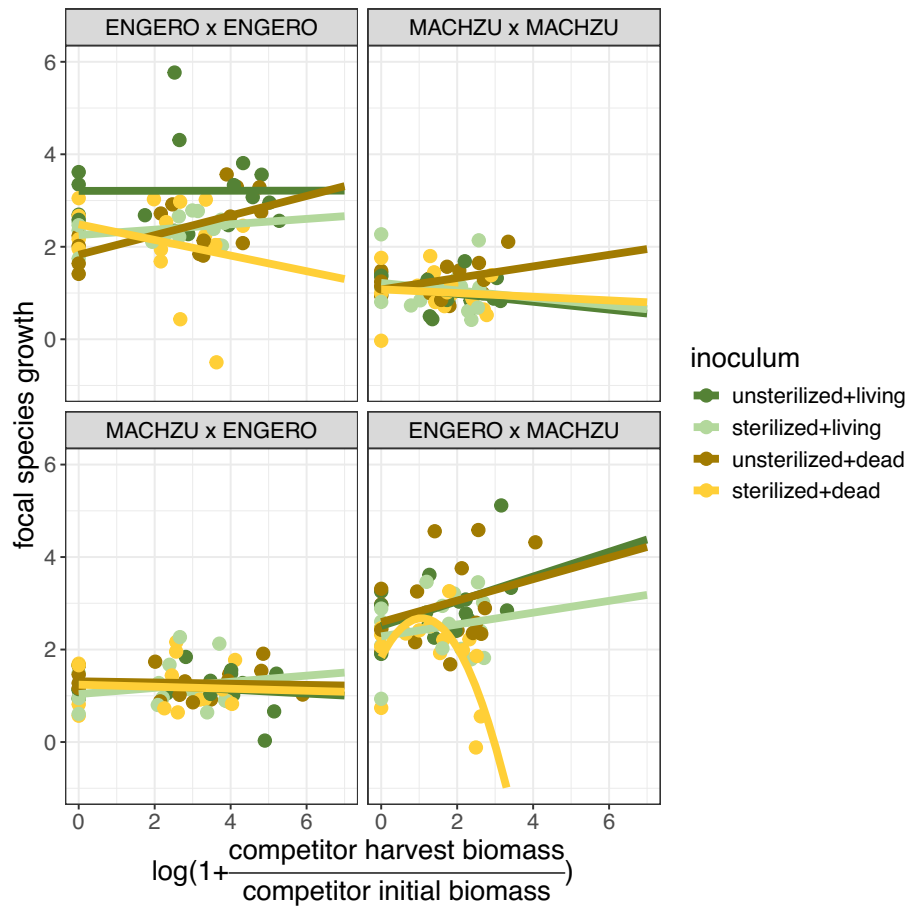


Fig. 2 Model fits depicting the effect of plant competitor biomass on the focal individual's growth. The linear model is fitted for each combination of competing species and inoculum type independently. Panel subtitles indicate the combination of competing species, where the first abbreviation indicates the focal species and the second its competitor (ENGERO = *Engelhardia roxburghiana* and MACHZU = *Machilus zuihoensis*). The competitor species is also the host of the soil inoculum. Inoculum type is a combination of sterilization treatment (unsterilized or sterilized) and host plant status (living or dead),

represented by different colors. The y-axis represents the focal individual growth, which is calculated as the log-ratio of its dry biomass at harvest, which is calculated as the log-ratio of its dry biomass at harvest to the average initial dry biomass of the species (i.e., $\log(\text{biomass}_{\text{harvest}}/\text{biomass}_{\text{initial}})$). The x-axis represents the log-transformed, dimensionless competitor biomass, defined as $\log(1 + \text{biomass}_{\text{harvest}}/\text{biomass}_{\text{initial}})$; scaling by initial seedling biomass avoids sensitivity to measurement units. Linear or quadratic regression lines are drawn depending on the model selection outcome

itive outcomes for each inoculum (indicated by solid triangles in Fig. 4a and b). Using the bootstrap samples, we assessed the robustness of species persistence and competitive outcomes. Based on the sign of IGRs and our 95% threshold criterion, we found robust inferences of species persistence in some cases. Specifically, for *E. roxburghiana*, 98.9% of bootstrap samples suggested a positive IGR in unsterilized inocula from living hosts, while 99.6% suggested a negative IGR in sterilized inocula from dead hosts (Table S5). For *M. zuihoensis*,

no IGRs met the 95% threshold to support a robust inference of persistence. We found notable uncertainty in the inferred competitive outcomes for all inoculum types (Table S6). Nevertheless, we still observed substantial differences in the bootstrap distributions of the inferred species persistence and competitive outcomes among sterilization treatments and host plant statuses.

By comparing species persistence and competitive outcomes in unsterilized versus sterilized inocula, we were able to detect how microbes mediated

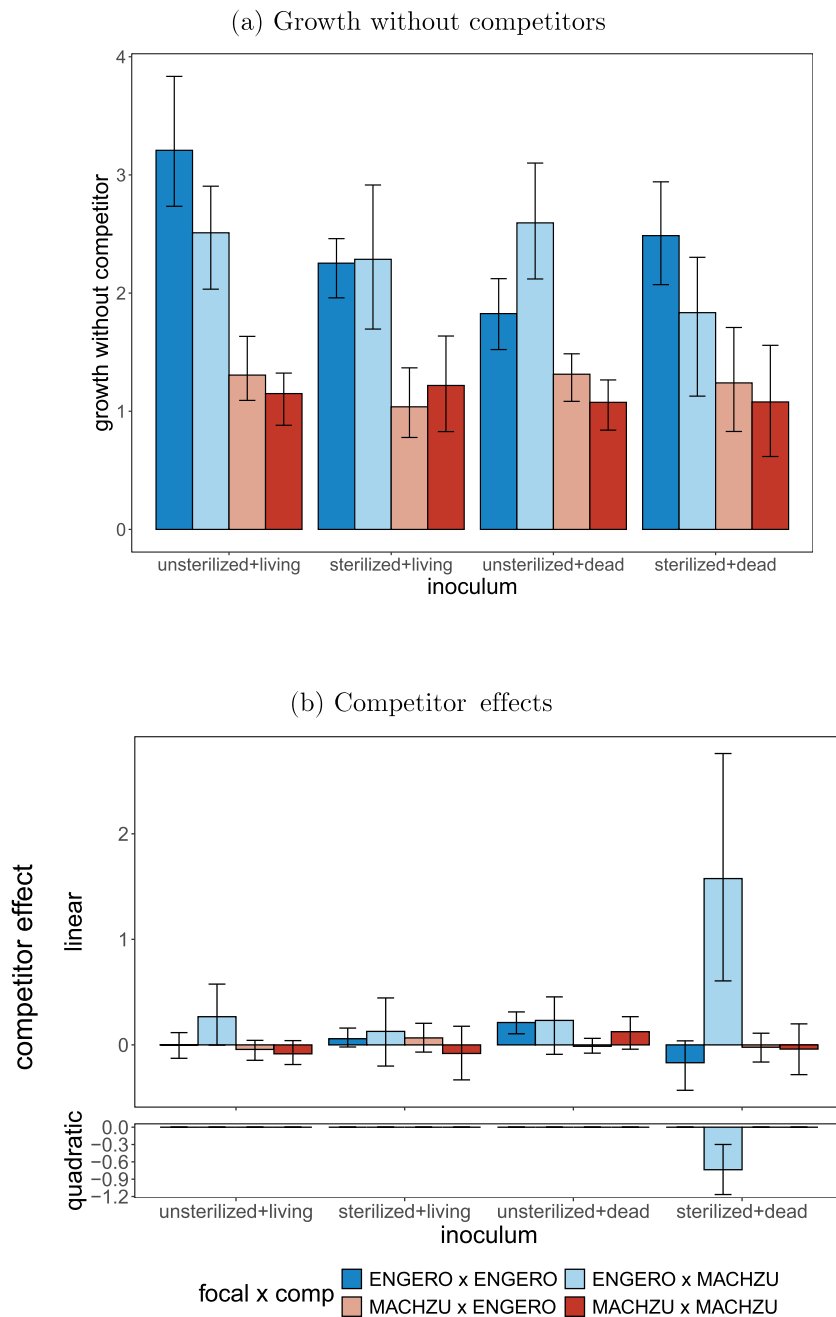


Fig. 3 Bootstrap distribution of (a) single-individual growth without competitors and (b) competitor effects for each inoculum type (combination of sterilization treatment and host tree statuses). Bar colors indicate the focal $\times$ competitor species combinations, where the competitor species is also the host of the soil inoculum (species acronyms: ENGERO = *Engelhardtia roxburghiana*; MACHZU = *Machilus zuihoensis*). Bar heights

represent (a) the estimated growth without competitors (intercepts) and (b) the estimated effects of competitor biomass on focal species' growth (linear and quadratic terms), derived from linear regressions of focal species' growth on transformed competitor biomass (Fig. 2). Error bars represent the 90% bootstrap confidence intervals for each model parameter

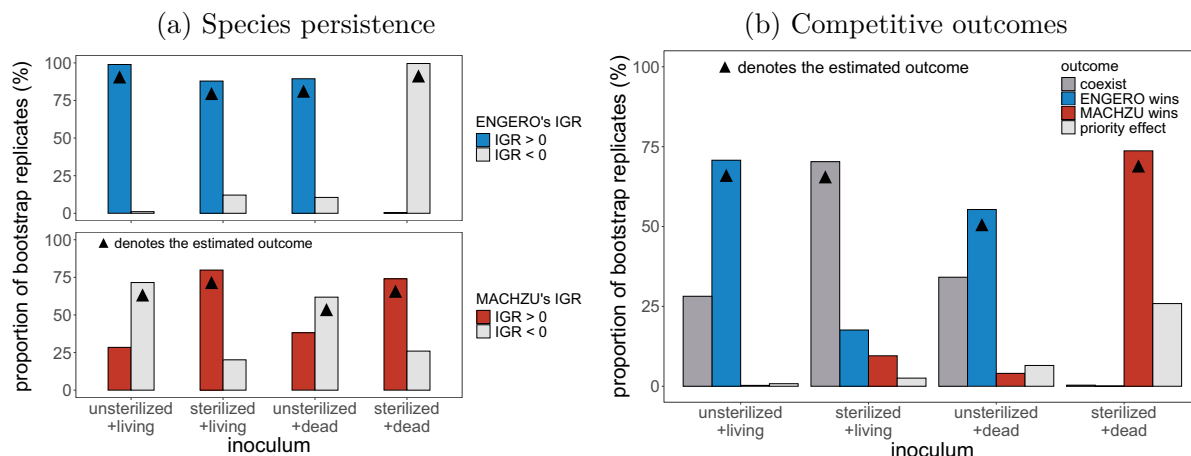


Fig. 4 Bootstrap distribution of (a) species persistence and (b) competitive outcome. Bar colors in (a) indicate the sign of the invasion growth rate (IGR) for the two species (blue: IGR > 0 for ENGERO = *Engelhardia roxburghiana*; red: IGR > 0 for MACHZU = *Machilus zuihoensis*; gray: IGR < 0). Bar colors in (b) indicate the competitive outcome (blue: ENGERO wins;

red: MACHZU wins; dark gray: coexistence; light gray: priority effect). Bar heights represent the proportion of bootstrap samples suggesting each IGR sign and competitive outcome. Black triangles indicate estimated persistence or coexistence outcomes, i.e., those obtained using the entire dataset as opposed to bootstrap resamples

seedling competition. Specifically, we found differences in species persistence and competitive outcomes between unsterilized and sterilized inocula collected from either living or dead hosts. For inocula conditioned by living hosts (first and second set of bars in Fig. 4a and b), the proportion of bootstrap samples suggesting *M. zuihoensis*'s persistence was lower in unsterilized inocula (i.e., decreased from 79.8% in sterilized inocula to 28.5% in unsterilized inocula; Table S5). This resulted in a lower proportion of coexistence (from 70.3% to 28.2%; Table S6) and a higher proportion of competitive wins by *E. roxburghiana* (from 17.6% to 70.8%) in the unsterilized inocula of living hosts. A similar pattern was observed in inocula collected from dead hosts (third and fourth set of bars in Fig. 4a and b). The proportion of bootstrap samples suggesting *M. zuihoensis*'s persistence was also lower in unsterilized inocula (i.e., decreased from 74.0% in sterilized inocula to 38.2% in unsterilized inocula; Table S5). However, unlike its consistently high persistence in inocula collected from living hosts, *E. roxburghiana*'s persistence was different among unsterilized and sterilized inocula collected from dead hosts (0.4% in sterilized inocula but 89.5% in unsterilized inocula). This shift resulted in a lower proportion of competitive wins by *M. zuihoensis* (from 73.7% to 4.0%; Table S6) but more wins by *E. roxburghiana* (from 0.1% to 55.3%) in the

unsterilized inocula of dead hosts. These differences between unsterilized and sterilized inocula suggest that soil microbes under both living and dead hosts mediated seedling competition, specifically by lowering the persistence of *M. zuihoensis* and enhancing the success of *E. roxburghiana*.

Furthermore, we determined the temporal change in species persistence and competitive outcomes by comparing treatments with inocula collected from living hosts in 2023 and from hosts that died between 2014 and 2019 (first and third set of bars in Fig. 4a and b). First, we found no clear difference in persistence and competitive outcomes between unsterilized inocula collected from living versus dead hosts: *E. roxburghiana* had a consistently higher proportion of persistence (98.9% in living host and 89.5% in dead host inocula; Table S5), while *M. zuihoensis* had a consistently lower proportion of persistence (28.5% in living host and 38.2% in dead host inocula). Accordingly, the unsterilized inocula from different host tree statuses resulted in similar bootstrap distributions of inferred competitive outcomes (Table S6). Digging deeper into the fitted model coefficients, recall that we found *E. roxburghiana* had a greater single individual growth in unsterilized inoculum from living hosts, compared to the one from dead hosts (Fig. 3a; Table S3). Moreover, its conspecific competitor effect shifted from non-

significant in inoculum from living hosts (estimate = 0.0008 with 90% CI: $-0.127 - 0.116$; Fig. 3b; Table S4) to significantly positive in inoculum from dead hosts. Despite these differences, the monoculture equilibrium of *E. roxburghiana* remained similar and its competitive impact on *M. zuihoensis* did not differ significantly across host statuses, resulting in a similar IGR bootstrap distribution for *M. zuihoensis* (see IGR calculation in “Data analysis” section). Together, these results suggest that microbial mediation of seedling competition remains consistent for 4–9 years after host death, despite detectable differences in single individual growth and competitor effects.

Finally, we found differences in species persistence and competitive outcomes between the sterilized inocula from different host statuses (second and fourth set of bars in Fig. 4a and b). Specifically, there was a higher proportion of bootstrap samples suggesting *E. roxburghiana*’s persistence in sterilized inocula from living hosts (87.9%), compared to those from dead hosts (0.4%; Table S5). This difference led to a shift in the bootstrap distribution of competitive outcomes: coexistence in 70.3% and *M. zuihoensis* wins in 9.6% of bootstrap samples in sterilized inocula from living hosts, compared to 0.4% and 73.7%, respectively, in sterilized inocula from dead hosts (Table S6). These results suggest that non-microbial legacies associated with host death can influence seedling competition in sterilized inocula.

Discussion

By combining field tree mortality data with a density-gradient greenhouse experiment, we revealed the important role of soil microbial communities in mediating seedling competitive outcomes under both living and dead trees. However, the hypothesis that the effects of PSF on seedling competition would vary through time was not supported by our results. Indeed, we found no differences in microbially mediated competitive outcomes over 4–9 years, as intact soil microbial communities consistently favored the exclusion of *M. zuihoensis* by *E. roxburghiana*, regardless of whether the inocula were sourced from living or dead trees. These temporally persistent PSFs align with Bennett et al. (2022), who showed similar consistent negative effects of soils from dead trees on conspecific seedlings’ survival and growth in a greenhouse experiment, and

with Magee et al. (2024), who showed that dead trees impose strong and pervasive negative density dependence on conspecific survival. Notably, a wealth of evidence suggests that key microbial regulators of plant performance, such as oomycete pathogens (Esch and Kobe 2021), arbuscular mycorrhizal fungi (Varga et al. 2015), and ectomycorrhizal fungi (Mueller et al. 2019, Shemesh et al. 2023), may persist for years to decades after host death through long-lived propagules, providing a mechanistic basis for temporally persistent PSF effects. Adding to these studies, our findings highlight the importance of soil microbial legacies in plant competition in real field systems.

Furthermore, our manipulative experiment suggested that the drivers of PSF may have shifted over time, even when the overall competitive consequences of soil effects were unchanged. In particular, sterilized inoculum treatments allowed us to isolate abiotic effects, i.e., those that did not depend on an active microbial community. In contrast to the unsterilized treatments where it always persisted, *E. roxburghiana* persisted in sterilized inoculum from living but not from dead hosts. We speculate that this shift in competitive outcomes with host plant death was driven primarily by abiotic factors. Despite the low proportion of inoculum (10%) in our soil treatments, plants responded to abiotic differences between the two sterilized inocula, likely reflecting strong nutrient sensitivity in our unfertilized pots or the potential role of sterilization-resistant allelochemicals. Indeed, during the first years after plant death, altered nutrient cycling may accelerate losses of key plant nutrients (Weis et al. 2006), while input and decomposition of dead tissue may elevate allelopathic effects (Xuan et al. 2005). However, in contrast to our findings in sterilized soils, evidence from unsterilized soils suggested that the intact microbial community can mask underlying abiotic differences, yielding similar competitive outcomes across living versus dead host soils. Accordingly, we hypothesize that the temporally persistent PSF observed in our study system, regardless of the change in the sterilized baseline, may reflect the interplay between symbiotic and abiotic factors: selection from a long-lived microbial propagule bank can maintain consistent PSF effects even as other processes, such as nutrient cycling, are altered. This interpretation accords with growing evidence that plants actively select distinct microbial communities (Duhamel et al. 2019) and that such microbial symbioses can main-

tain plant performance under changing stressors such as nutrient availability (Bogar et al. 2022) and competition (Willing et al. 2024). More broadly, we argue that the strong abiotic differences we observed likely still reflect the indirect outcome of microbe-mediated processes such as decomposition, nutrient cycling (Weis et al. 2006), and the chemical transformation of allelochemicals (Cipollini et al. 2012). By demonstrating the interplay between microbial and abiotic factors, our results add to recent efforts untangling how biotic, abiotic, and interactive effects contribute to plant legacies (Dudenhöffer and Hulme 2025).

Indeed, in line with recent calls (in 't Zandt et al. 2020) to address soil nutrient availability as a mechanism mediating plant–soil microbe interactions, Van Nuland et al. (2023) combined competition, nitrogen addition, and inoculum sterilization treatments to show how context-dependent interactions between soil nutrients and plant–microbe symbioses jointly determine competitive outcomes between tree species. Our findings add an important temporal dimension to this line of research: persistence of microbial communities after plant death may interact with concurrent changes such as altered nutrient cycling to determine the overall impact of PSFs on seedling competition. Pinpointing the plant and microbial processes responsible for persistent soil legacies will require experiments incorporating microbial community changes after host death (Ou et al. 2024), in tandem with diverse mechanistic techniques such as nutrient addition (Van Nuland et al. 2023), nutrient tracing (Bogar et al. 2022), or molecular methods including genomics (Janse van Rensburg et al. 2025), transcriptomics (Donnelly et al. 2026), and metabolomics (Semchenko et al. 2026). Moreover, the distribution of key microbial symbionts (e.g., ectomycorrhizal fungi) is known to vary within single trees' rooting zones (Dickie and Reich 2005, Peay et al. 2011) as well as between individuals across the landscape (Gao et al. 2015). Thus, we point to the vital need to place legacy effects within an explicitly spatial context (Ke et al. 2025) through designs with replicates across different distances or individuals (reviewed in Chung 2023, Gundale et al. 2017).

Our findings also indicate important species-level differences in PSF and its consequences over time. We always detected greater persistence of *E. roxburghiana* in unsterilized soil versus sterilized controls, an effect that was particularly strong in soils from dead hosts. This pattern suggests that *E. roxburghiana* benefited

from soil microbes, likely due to ectomycorrhizal symbioses providing nutrient benefits (Segnitz et al. 2020) or pathogen protection (Bennett et al. 2017). These persistent positive feedbacks, writ large in the context of a multispecies community (Eppinga et al. 2018), may contribute to the high abundance of *E. roxburghiana* at Fushan FDP, as shown for other dominant tree species in species-rich forest ecosystems (Segnitz et al. 2022). On the other hand, our experiment provided stronger support for the persistence of *M. zuihoensis* in sterilized than in unsterilized inocula. This suggests that *M. zuihoensis* experienced negative microbial effects. Though we evaluated only one species pair in the study, previous comparisons showing that arbuscular mycorrhizal plants tend to show more negative PSFs than their ectomycorrhizal competitors may provide one possible explanation for this species-level difference in PSF (Bennett et al. 2017, Eagar et al. 2025). Moreover, host death had little impact on bootstrap support for *M. zuihoensis* persistence, regardless of whether the inoculum was sterilized. Nor did we detect any significant changes in the slope or intercept of competition models for *M. zuihoensis* between living versus dead host soils, in contrast to the treatment-specific differences for *E. roxburghiana*. We hypothesize that the consistent negative microbial effects experienced by *M. zuihoensis* may be driven by its close phylogenetic relationship to surrounding vegetation, reflecting the high abundance of Lauraceae combined with the lack of non-random phylogenetic neighbor clustering reported at Fushan FDP (Wang et al. 2016). In other words, soil conditioning by abundant confamilial neighbors may maintain negative soil microbial effects for *M. zuihoensis* seedlings even after adult hosts die. This parallels results from Segnitz et al. (2020), who found that arbuscular mycorrhizal trees in a hyperdiverse tropical forest experience negative PSF from close relatives, and emphasizes pathogen spillover from common species as a key mechanism in maintaining microbial effects after plant death. Indeed, previous work at Fushan FDP has shown that trees tend to suffer stronger negative density-dependent effects from more closely related species (Wills et al. 2021). More broadly, our findings point to the importance of further mechanistic work on how PSF persistence varies between host plants, especially in species-rich low-latitude forests where symbiotic traits and phylogenetic relatedness are known to strongly modulate PSFs (Liang et al. 2020, Segnitz et al. 2020, Surendra et al. 2026).

Using a density gradient, we untangled the nuanced impacts of soil microbes on different aspects of competition. Importantly, complex changes in density dependence were observed for *E. roxburghiana*: we found that soil microbial effects from living trees increased *E. roxburghiana*'s growth alone, while microbial legacies from dead trees resulted in positive (i.e., facilitative) intraspecific density dependence for *E. roxburghiana*. Although these microbial effects influenced *E. roxburghiana* in different manners, both ultimately promoted its dominance during seedling competition when it imposed negative competitor effects on *M. zuihoensis*. Large-scale studies on legacy effects have likewise shown changes in density-dependent impacts on demographic rates (e.g., Magee et al. 2024). Thus, we underscore the need to incorporate density-dependent microbial effects, long studied in the context of plant population ecology (Watkinson and Freckleton 1996), into experimental PSF studies. The experimental design we apply here (Ke and Wan 2023) provides a useful bridge between these two approaches, especially in systems such as ours, where conditioning involves much larger individuals and longer timescales than experimental responses. Meanwhile, model selection identified significant nonlinearity in the response of *E. roxburghiana* competing with *M. zuihoensis* in sterilized soil collected from dead trees (Fig. 3b). Moreover, a significant positive linear effect was observed in *E. roxburghiana*'s conspecific competition in unsterilized dead host soils. As biological realism dictates that performance cannot increase indefinitely with competitor density (Ke and Wan 2023), this result provides indirect evidence for an unobserved nonlinear response. These patterns might indicate the role of microbial communities in shaping the nonlinearity of seedling competition (as found by West 1996, Willing et al. 2024). As theoretical work has already demonstrated that complex demographic responses may arise from the underlying biology of species interactions (e.g., Holland and DeAngelis 2010, Letten and Stouffer 2019), studying these nonlinearities is especially important in light of the long-standing debate regarding appropriate functional forms for competitive responses (Ayala et al. 1973) and their coexistence implications (Armitage 2024, Hatton et al. 2024). We emphasize the utility of density gradient experiments in providing a more complete picture to link PSF mechanisms to realistic, potentially nonlinear, competition functional responses.

Nonetheless, our findings also highlight methodological challenges in quantifying density-dependent microbial effects. Similar to previous studies quantifying plant coexistence (e.g., Chu and Adler 2015), including those from the perspective of plant–soil interactions (Willing et al. 2024), we found that estimated competitive effects were close to zero in many cases, contributing to considerable uncertainty in the predicted competitive outcomes. This likely reflects the difficulty of choosing density treatments that allow accurate estimation of monoculture equilibrium (Ke and Wan 2023). In other cases, challenges in model-fitting may be related to potential nonlinearities in competitive responses. For instance, though one nonlinear term was retained and bootstrapping indicated robust support for some positive competitive responses (Fig. 3b), parameter uncertainty was relatively large. While we were still able to apply our invasion analysis-based approach to qualitatively predict coexistence, our results emphasize the importance of assessing model robustness. Accordingly, we echo recent recommendations to assess uncertainty via model selection (Armitage 2024) and bootstrapping (as implemented here; Fig. 4), or other techniques including error propagation (Yan et al. 2022) or Bayesian statistics (Terry and Armitage 2024).

In summary, by combining long-term field data with a density-gradient greenhouse experiment, we showed that microbial effects can persist for 4–9 years after host death and consistently affect seedling competitive outcome. Our results also highlight the potential for interactions between consistent microbial communities and temporal shifts in abiotic conditions. Moreover, we revealed the role of soil microbes in mediating the nonlinear density dependence in seedling competitive responses. Crucially, these microbial effects act through density-dependent competition functions, which highlights that assessing plant–soil interactions using only single individuals may overlook important features of PSF. Looking forward, we recommend combining density gradient experiments with robust methods for model selection and uncertainty estimation. Moreover, future work may gain further mechanistic insight by pairing density dependent plant–soil feedback experiments with direct measurements of microbial turnover. Together, these approaches will improve our ability to predict plant community dynamics through the lens of plant–soil microbe interactions.

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Data Availability All data and computer scripts supporting the results are archived at the following public repository: <https://doi.org/10.5281/zenodo.20875201>.

Declarations

Competing interests The authors declare no competing interests.

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